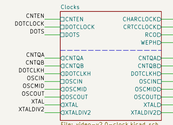
The video card V2.0, one sheet per functional block:
  sheet 2 Bus and ports BUS INTERFACE: PORT DECODE, CONTROL AND STATUS (PA-PE)
  sheet 3 CRTC   6845 CRTC ON PORTS PA (ADDRESS) / PB (DATA)
  sheet 4 Text   BOARD SELECT ($0000 BLOCK), TEXT RAM $0000-$07FF, CHARACTER GENERATOR, TEXT DOT SHIFTER
  sheet 5 Clocks   DOT CLOCK, CHARACTER COUNTER (DOTS: 6 OR 8 DOTS A CHARACTER)
  sheet 6 Graphics   GRAPHICS RAM $0800-$0FFF: SLOT MODE (GFX-ON = 1) / DIRECT MODE (GFX-ON = 0)
  sheet 7 Video out   DOT COMBINE AND VIDEO OUT (GRAPHICS ONLY IN 8-DOT MODE)
  sheet 8 PS2   PS/2: ATMEGA328P. LATCHES ON PD / PE. FTDI. ~INT. RESET
  sheet 9 Power   POWER UNITS, DECOUPLING, POWER LED, SPARE GATES

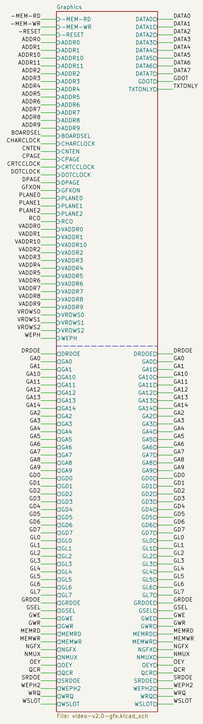
Every net of a sheet is a pin of its sheet symbol here, and the local label at each pin names the net
(/NAME, the names on the board). Above the dashed line: the nets the sheet shares with other sheets (left:
it reads them; right: it drives them). Below it: the nets drawn on that sheet only, a pin on each side
(KiCad's ERC counts a root label as connected only when another label of its name is on the root sheet).
VCC and GND are power symbols on the sheets.



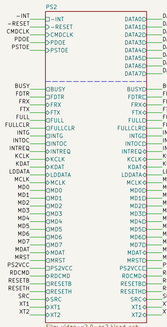












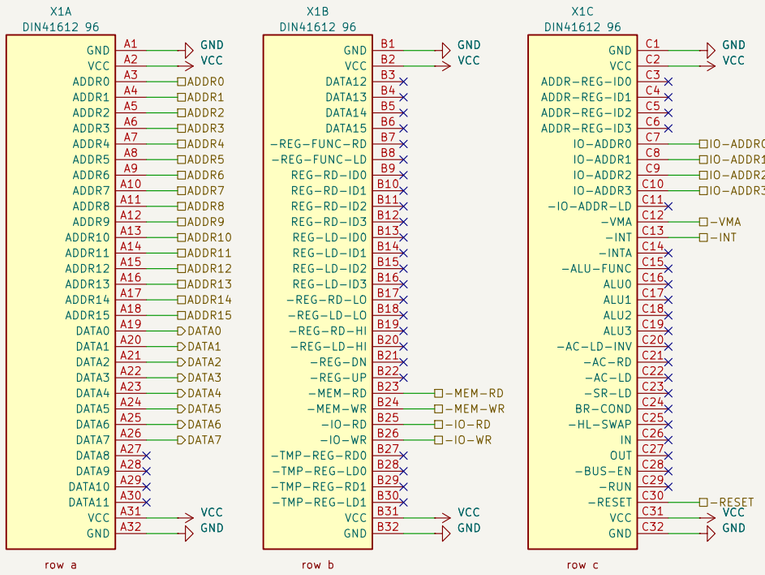


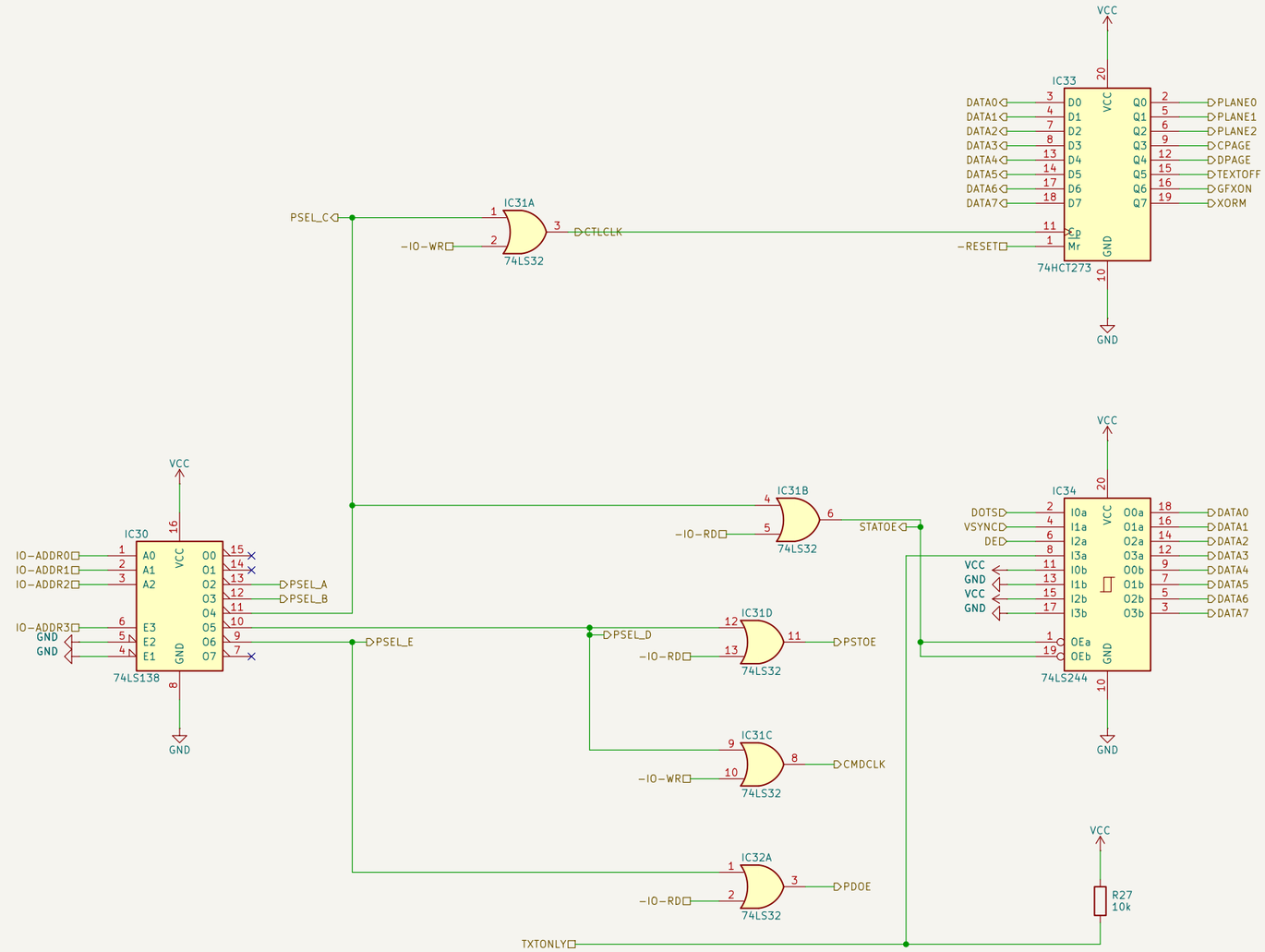
Design note: BACKLOG.md Video card V2.0: theory: docs/cards/video.md
Circuit source: v20_netlist.py: these sheets are generated by gen_v20.py
Text + graphics (8-dot mode), 6845 and every register on ports PA-PE, PS/2 keyboard and mouse

YACC1	
Sheet: /	
File: video-v2.0.kicad_sch	
Title: YACC1 video card V2.0	
Size: A1	Date: 2026-10-23
KiCad E.D.A. 10.0.6	**Rev: V2.0**
	Id: 1/9

BUS INTERFACE, PORT DECODE, CONTROL AND STATUS (PA–PE)

X1: the YACC1 V3.2 bus (DIN 41612). IC30 decodes IO-ADDR0-3: Y2..Y7 = ports PA..PF when IO-ADDR3 = 1.
IC31 / IC32 gate the strobes: PC write = control register IC33 (cleared by -RESET), PC read = status IC34
(DOTS, VSYNC, DE, TEXT-ONLY, card ID 0101), PD write / read = the PS/2 command latch / status, PE read = PS/2 data.





Design note: BACKLOG.md Video card V2.0; theory: docs/cards/video.md
Circuit source: v20_netlist.py; this sheet is generated by gen_v20.py
Sheet 2 of the video card V2.0 schematic (root: video-v2.0.kicad_sch)
YACC1

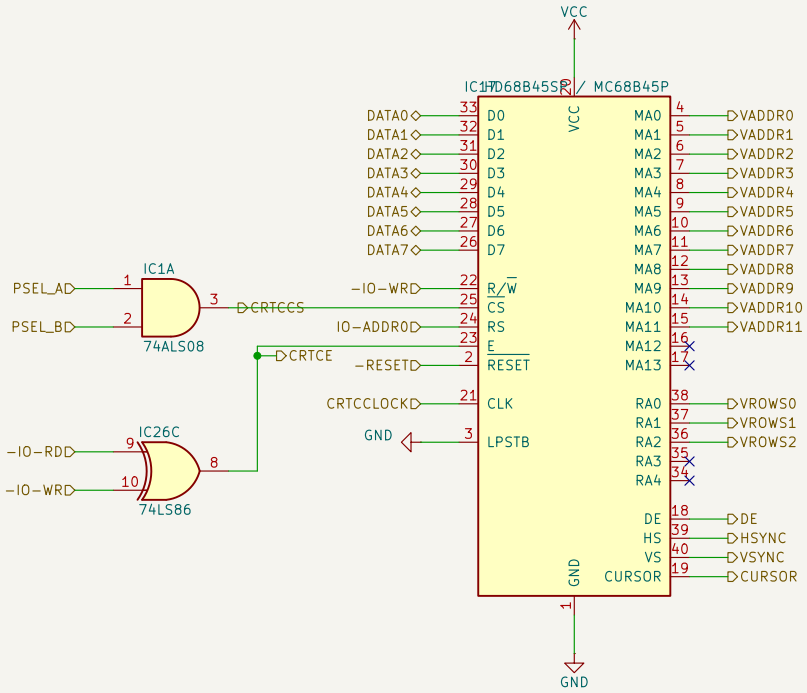
Sheet: /Bus and ports/
File: video-v2.0-bus.kicad_sch

Title: YACC1 video card V2.0: Bus and ports

Size: A3	Date: 2026-10-23	**Rev: V2.0**
KiCad E.D.A. 10.0.6		Id: 2/9

6845 CRTC ON PORTS PA (ADDRESS) / PB (DATA)

−CS = PSEL_A AND PSEL_B (IC1/A): low for PA or PB. RS = IO−ADDR0, R/−W = −IO−WR,
E = −IO−RD XOR −IO−WR (IC26/C): high during either strobe. MA0−11 and RA0−2 address the text RAM, the
character generator and the graphics muxes; the 6845 grade: README (the B grade above 1 MHz).



Design note: BACKLOG.md Video card V2.0; theory: docs/cards/video.md
Circuit source: v20_netlist.py; this sheet is generated by gen_v20.py
Sheet 3 of the video card V2.0 schematic (root: video−v2.0.kicad_sch)

YACC1

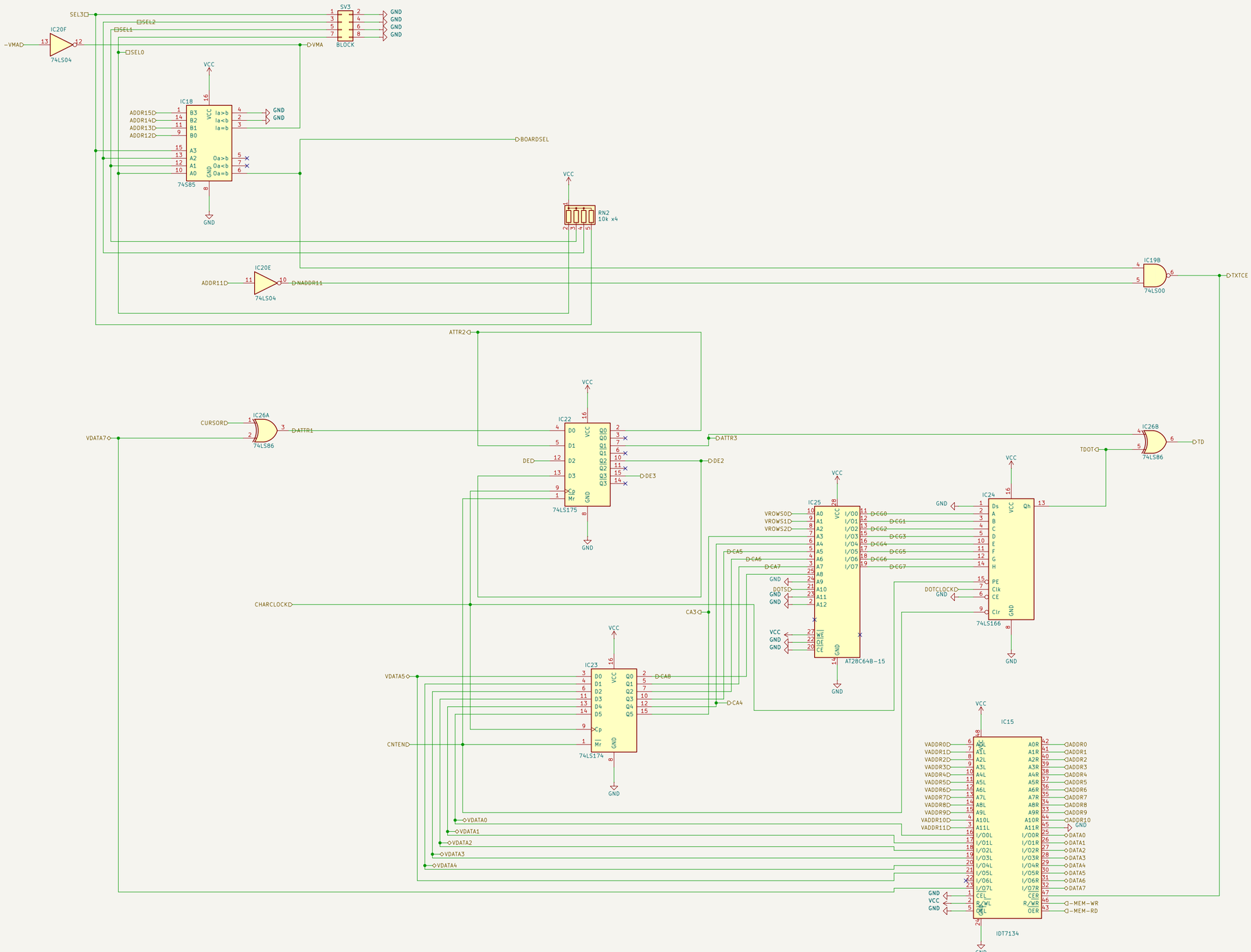
Sheet: /CRTC/
File: video−v2.0−crtc.kicad_sch

Title: YACC1 video card V2.0: CRTC

Size: A4	Date: 2026−10−23	**Rev: V2.0**
KiCad E.D.A. 10.0.6		Id: 3/9

BOARD SELECT ($D000 BLOCK), TEXT RAM $D000–$D7FF, CHARACTER GENERATOR, TEXT DOT SHIFTER

IC18 compares ADDR12–15 with SV3 (fitted = 0; 5–6 only = $D000) while VMA: BOARDSEL. The CPU side of the IDT7134 is $D000–$D7FF (ADDR11 = 0, IC19/B): the 6845 reads the other port. IC23 latches the code, IC25 (28C64, A10 = DOTS: the font) gives the row, IC24 shifts it out; IC22 delays DE and the attribute bits.



Design note: BACKLOG.md Video card V2.0; theory: docs/cards/video.md
Circuit source: v20_netlist.py: this sheet is generated by gen_v20.py
Sheet 4 of the video card V2.0 schematic (root: video-v2.0.kicad_sch)
YACC1

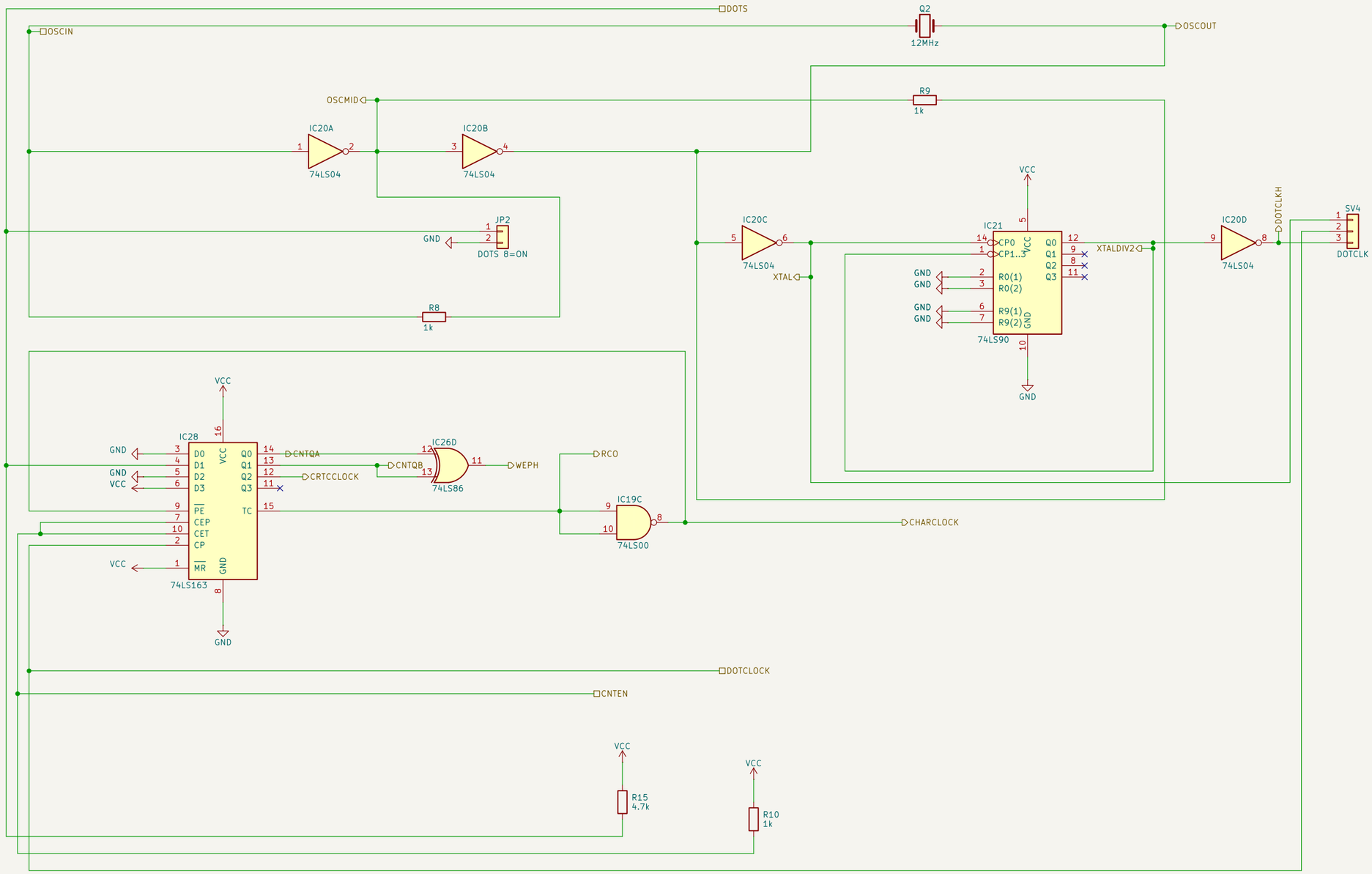
Sheet: /Text/
File: video-v2.0-text.kicad_sch

Title: YACC1 video card V2.0: Text

Size: A2	Date: 2026-10-23	**Rev: V2.0**
KiCad E.D.A. 10.0.6		Id: 4/9

DOT CLOCK, CHARACTER COUNTER (DOTS: 6 OR 8 DOTS A CHARACTER)

12 MHz crystal oscillator (IC20/A–C); SV4 1–2 = the crystal, 2–3 = half of it (IC21). IC28 counts the dots of a character and loads 8 (JP2 fitted, 8 dots) or 10 (open, 6 dots) on its carry: CHARCLOCK = NOT RCO (IC19/C). WEPH = QA XOR QB (IC26/D): counts 9 and 10, the graphics write phase.



Design note: BACKLOG.md Video card V2.0; theory: docs/cards/video.md
Circuit source: v20_netlist.py; this sheet is generated by gen_v20.py
Sheet 5 of the video card V2.0 schematic (root: video-v2.0.kicad_sch)
YACC1

Sheet: /Clocks/
File: video-v2.0-clock.kicad_sch

Title: YACC1 video card V2.0: Clocks

Size: A3	Date: 2026-10-23	**Rev: V2.0**
KiCad E.D.A. 10.0.6		Id: 5/9

GRAPHICS RAM $D800–$DFFF: SLOT MODE (GFX-ON = 1) / DIRECT MODE (GFX-ON = 0)

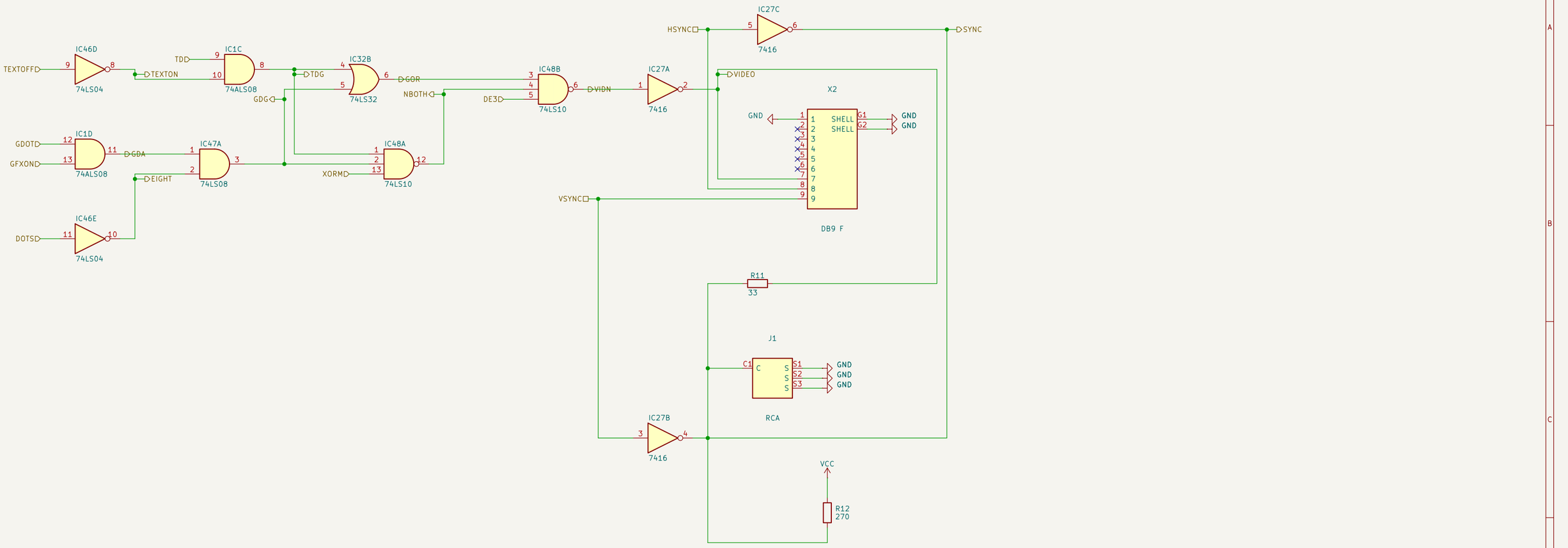
NMUX = NAND(QC OR RCO, GFX-ON): 1 = the CPU address on the muxes: it side. 0 = the display (MA, RA). Slot mode: WRQ (IC44/A) holds a CPU write for the next slot. ~WE = NAND(WRQ, NMUX, WEPH OR NOT GFX-ON, GWR), reads through the latch IC41; Direct mode: WRQ preset, ~WE = NOT GWR, reads through IC55. R25 / R26: the text-only build only (README). IC44/B is held cleared: its Q = IN PC bit 3 (0 = the graphics chips are fitted).

CRTCCLOCK
IC44B
74LS74
TXTONLY
IC35
74LS157
VADDR0D
VADDR1D
VADDR2D
VADDR3D
GA0
GA1
GA2
GA3
NMUX
IC36
VADDR4D
VADDR5D
VADDR6D
VADDR7D
GA4
GA5
GA6
GA7
GA8
IC37
VADDR8D
VADDR9D
VADDR10D
VROWS0D
PLANE0D
GA9
GA10
GA11
GA12
IC38
VROWS1D
PLANE1D
VROWS2D
PLANE2D
DPAGED
CPAGED
GA13
IC39
62256
GA14
SRDOE
GD0
GD1
GD2
GD3
GD4
GD5
GD6
GD7
RN3
10k x8
IC41
74HC574
DATA0
DATA1
DATA2
DATA3
DATA4
DATA5
DATA6
DATA7
IC42
74LS374
GL0
GL1
GL2
GL3
GL4
GL5
GL6
GL7
IC43
74LS166
DOTCLOCKD
CNTEN
R25
1k
GDOT
IC32C
74LS32
RCO
DCR
IC19D
74LS00
IC55
74HCT244
GRDOE
IC54A
74HCT08
GSEL
IC47B
74LS08
BOARDSEL
ADDR11D
IC19A
74LS00
MEMRD
IC54B
74LS20
VCC
NGFX
IC1B
74LS08
WRQ
WSLOT
GWE
IC40
74HCT244
IC48C
74LS10
IC46B
74LS04
-MEM-RDD
GFXON
IC47C
74LS08
GWR
IC46C
74LS04
-MEM-WRD
-MEMWR
IC45A
74LS20
WEPH2
CHARCLOCKD
IC46A
74LS04
IC44A
74LS74
-RESETD-
R26
1k
IC32D
74LS32
WEPHD

Design note: BACKLOG.md Video card V2.0: theory: docs/cards/video.md
Circuit source: v20_netlist.py: this sheet is generated by gen_v20.py
Sheet 6 of the video card V2.0 schematic (root: video-v2.0.kicad_sch)
YACC1
Sheet: /Graphics/
File: video-v2.0-gfx.kicad_sch
Title: YACC1 video card V2.0: Graphics
Size: A1 Date: 2026-10-23 Rev: V2.0
KiCad E.D.A. 10.0.6 Id: 6/9

DOT COMBINE AND VIDEO OUT (GRAPHICS ONLY IN 8-DOT MODE)

Text dot TD AND TEXT-ON, graphics dot GDOT AND GFX-ON AND EIGHT; TEXT OR GRAPHICS, or with XOR set the exclusive OR (IC48/A,B, IC32/B); gated by DE3 (IC48/B). IC27 (7416, open collector) drives the DB9 TTL video (7 video, 8 HSYNC, 9 VSYNC) and the composite RCA (video into the sync node R11 / R12).



Design note: BACKLOG.md Video card V2.0; theory: docs/cards/video.md
Circuit source: v20_netlist.py; this sheet is generated by gen_v20.py
Sheet 7 of the video card V2.0 schematic (root: video-v2.0.kicad_sch)
YACC1

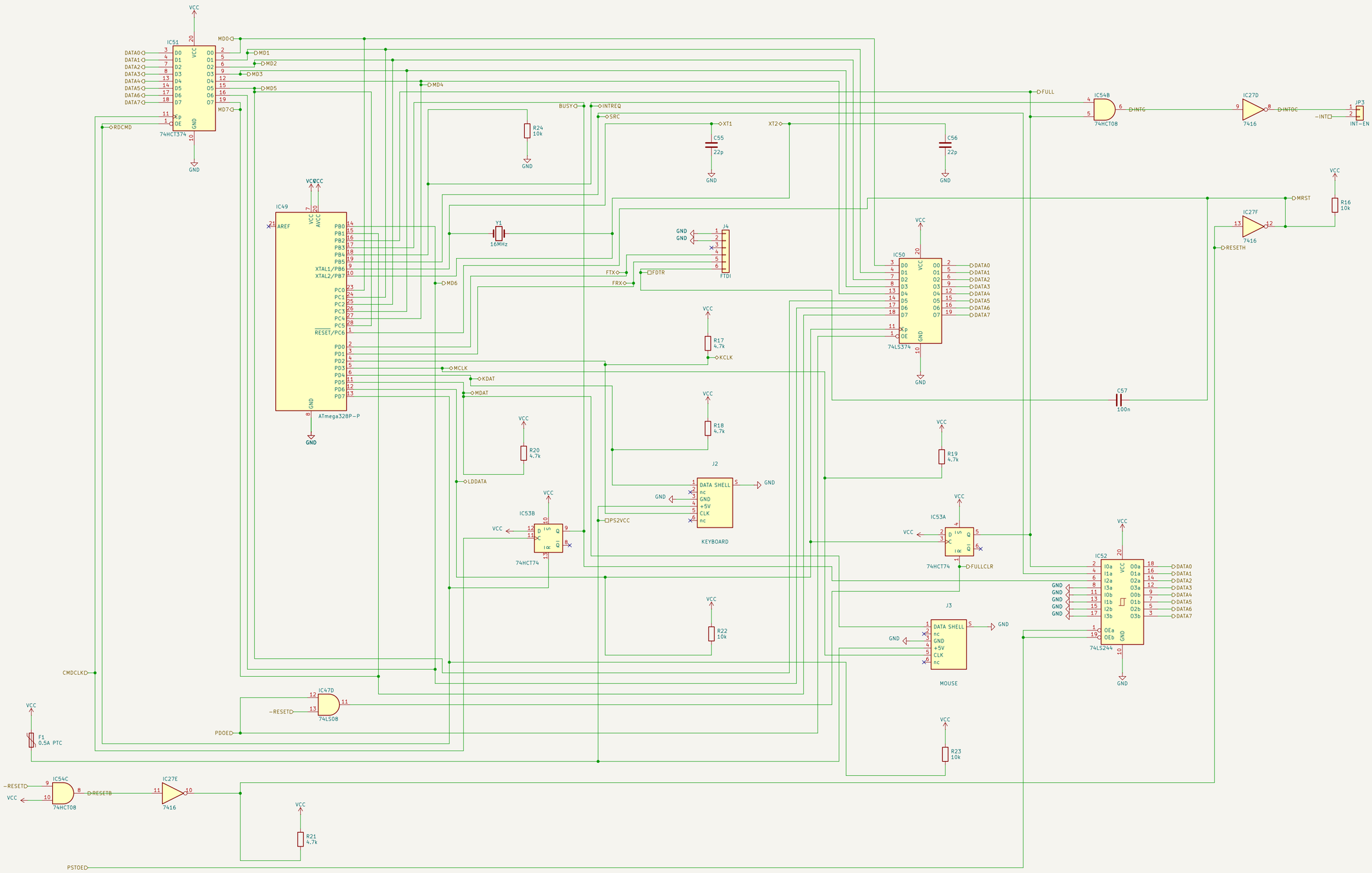
Sheet: /Video out/
File: video-v2.0-video.kicad_sch

Title: YACC1 video card V2.0: Video out

Size: A3	Date: 2026-10-23	**Rev: V2.0**
KiCad E.D.A. 10.0.6		Id: 7/9

PS/2: ATMEGA328P, LATCHES ON PD / PE, FTDI, −INT, RESET

IC49 runs embedded/video−ps2. OUT PD loads a command into IC51 (BUSY set), IN PE reads a byte from IC50 (FULL cleared). IN PD the status IC52. −INT = INTREQ AND FULL (IC54/B) through IC27/D and JP3 (INT−EN). The bus −RESET reaches MRST through IC54/C, IC27/E and IC27/F (open collector, R21): it only ever pulls MRST low.



Design note: BACKLOG.md Video card V2.0; theory: docs/cards/video.md
Circuit source: v20_netlist.py; this sheet is generated by gen_v20.py
Sheet 8 of the video card V2.0 schematic (root: video−v2.0.kicad_sch)

YACC1

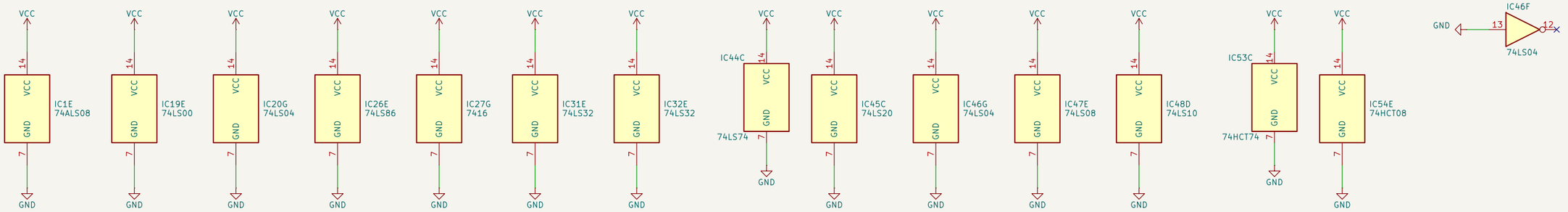
Sheet: /PS2/
File: video−v2.0−ps2.kicad_sch

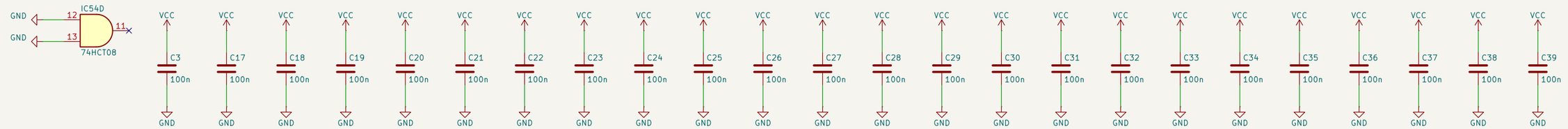
Title: YACC1 video card V2.0: PS2

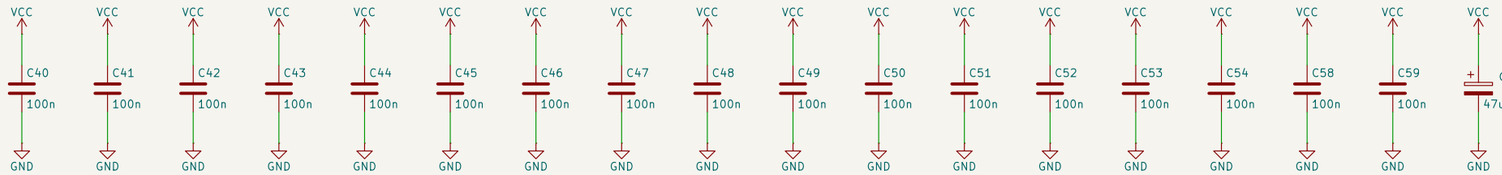
Size: A2	Date: 2026−10−23	**Rev: V2.0**
KiCad E.D.A. 10.0.6		Id: 8/9

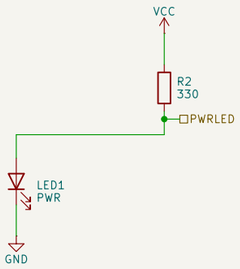
POWER UNITS, DECOUPLING, POWER LED, SPARE GATES

One 100 nF per IC beside its VCC pin (C54 at the ATmega's AVCC), C16 47 uF bulk. The bus supplies VCC and GND (X1, six pins each): the PWR_FLAGs. Spare: IC46/F (input on GND), IC54/D (inputs on GND), IC38's fourth section.











PWR_FLAGs: the bus supplies VCC and GND (X1, six pins each).

Design note: BACKLOG.md Video card V2.0; theory: docs/cards/video.md
Circuit source: v20_netlist.py; this sheet is generated by gen_v20.py
Sheet 9 of the video card V2.0 schematic (root: video-v2.0.kicad_sch)
YACC1

Sheet: /Power/
File: video-v2.0-power.kicad_sch

Title: YACC1 video card V2.0: Power

Size: A3	Date: 2026-10-23	**Rev: V2.0**
KiCad E.D.A. 10.0.6		Id: 9/9